

NETWORKS AND COMPLEXITY

Solution 21-6

*This is an example solution from the forthcoming book Networks and Complexity.
Find more exercises at <https://github.com/NC-Book/NCB>*

Ex 21.6: Laplacian as second derivative [Lvl. 3]

Consider diffusion in a one-dimensional continuous space that is the interval $[0, 5]$. In continuous space diffusion is described by the Laplace equation

$$\dot{x} = \Delta x = \frac{\partial^2}{\partial l^2} x$$

- a) Discretize the interval into five bins, numbered 1 to 5. Each bin i has an associated variable x_i that counts the number of walkers between $l = i - 1$ and $l = i$. Show that the dynamics in bin 3 is approximated by

$$\dot{x}_3 = -2x_3 + x_2 + x_4.$$

Solution

To discretize the spatial derivative we can consider a small change in l that is actually one unit long

$$dl = 1. \tag{1}$$

The corresponding change in $x(l)$ is

$$dx(l) = x(l + 0.5) - x(l - 0.5). \tag{2}$$

We can now approximate the derivative with these finite differences,

$$\frac{dx}{dl}(l) = x(l + 0.5) - x(l - 0.5). \tag{3}$$

For the second derivative, we need to compute a change in the first derivative if we move by one unit,

$$d \frac{dx}{dl}(l') = \frac{dx}{dl}_{l=l'+0.5} - \frac{dx}{dl}_{l=l'-0.5} \tag{4}$$

$$= x(l' + 1) - x(l') - x(l') + x(l' - 1) \tag{5}$$

$$= -2x(l') + x(l' + 1) + x(l' - 1). \tag{6}$$

Hence, we can write

$$\frac{d^2x}{dl^2}(l) = \frac{d}{dl} \frac{dx}{dl}(l) = -2x(l) + x(l + 1) + x(l - 1) \tag{7}$$

So, if we consider this at the center of bin 3, which is at the point $l = 2.5$

$$\frac{d^2x}{dl^2}(2.5) = -2x(2.5) + x(3.5) + x(1.5). \tag{8}$$

The diffusion equation tells us that the second spatial derivative equals the temporal derivative and hence

$$\dot{x}(2.5) = \frac{d^2x}{dl^2}(2.5) = -2x(2.5) + x(3.5) + x(1.5) \quad (9)$$

If we now discretize into bins the density of walkers $x(2.5)$ becomes the walker density in bin 3, which is x_3 . Likewise, $x(1.5)$ becomes the walker density in bin 2 (x_2) and $x(3.5)$ becomes the walker density in bin 4 (x_4). And hence, the change in x_3 is

$$\dot{x}_3 = -2x_3 + x_2 + x_4. \quad (10)$$

- b) Compare this result with the random walk on a network that consists of a chain of five nodes, numbered again from 1 to 5. What is the differential equation for x_3 , the number of walkers in node 3 in this model?

Solution

We know the random walk follows

$$\dot{\mathbf{x}} = -\mathbf{L}\mathbf{x} \quad (11)$$

The adjacency matrix for the 5-chain is

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix} \quad (12)$$

The degree matrix is

$$\mathbf{K} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \quad (13)$$

and hence the Laplacian is

$$\mathbf{L} = \mathbf{K} - \mathbf{A} = \begin{pmatrix} 1 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & 1 \end{pmatrix}. \quad (14)$$

We can write the i -th row of $\dot{\mathbf{x}} = -\mathbf{L}\mathbf{x}$ as

$$\dot{x}_i = -\sum L_{ij}x_j \quad (15)$$

and hence for node 3 we have the differential equation

$$\dot{x}_3 = -2x_3 + x_2 + x_4 \quad (16)$$

which is the same equation as before. Generally, it is a good intuition to think of the $-\mathbf{L}$ as the network version of the second spatial derivative.